

A predictive model of weld bead cross-section morphology during laser deep penetration welding considering energy coupling law in keyhole

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ABSTRACT

It is essential to develop an accurate and fast-solving prediction model of weld bead cross-section (WBCS) morphology to select appropriate processing parameters for industrial applications of laser deep penetration welding (LDPW). In this paper, considering laser multiple reflections and absorption within the keyhole and energy balance of the keyhole central section in the direction perpendicular to welding velocity, the mathematical relationship between the fusion boundary of WBCS and the keyhole boundary was obtained. Subsequently, a predictive model of WBCS morphology during LDPW was established using the above relationship and the keyhole boundary profile calculated by the morphology-calculated model for front keyhole wall (FKW) considered the primary absorption of FKW. The results of calculations indicated that the variation of WBCS morphology with laser power, welding velocity, defocusing amount, laser wavelength, and other parameters was largely consistent with the consensus. The model feasibility was verified by comparing the results from calculations and experiments. It can be found that the keyhole depth can be obtained by calculating based on the primary absorption of FKW. However, the laser energy coupling pattern within the keyhole is characterized by multiple reflections, mainly converging towards the keyhole bottom. Therefore, the characteristics of laser energy coupling in the keyhole should be comprehensively considered while calculating the WBCS morphology during LDPW.

1. Introduction

The technology of LDPW has been widely applied extensively in various industries, including aviation, transit, electronic information, and other manufacturing industries due to the advantages such as high processing efficiency, excellent joint quality, and processing flexibility [1–3]. The WBCS morphology is one of the characteristic features to measure the quality of welded joints. It involves a series of complex steps, including welding experiments, post-welding treatments, and metallographic processing, to obtain the morphology of WBCS under different welding process parameters through experimental means. However, this process can be resource-intensive, requiring significant workforce, materials, and financial investment. Therefore, establishing a precise mathematical model to predict WBCS morphology under different parameters is crucial for advancing the practical application of LDPW technology in engineering and understanding the welding physical process.

The morphology prediction of WBCS was achieved mainly by calculating temperature field distribution, and the WBCS morphology was obtained according to the calculated temperature field distribution [4,5]. For example, Chukkan et al. [6] calculated temperature field distribution during the pulsed Nd:YAG laser welding of 316 L stainless steel sheet by employing various three-dimensional heat sources, including a conical heat source, a combined source of double-ellipsoid and conical source, and a combined source of cylindrical and conical source, respectively. Kazemi et al. [7] realized the prediction of temperature field and WBCS morphology during laser full penetration welding utilizing a combined source consisting of a Gaussian disk source on the surface and a line source through depth. Xia et al. [8] predicted weld shape using the combined source of a Gaussian surface source and a cylindrical source with a damped peak value through depth during fiber laser welding, achieving a prediction error of less than 8.9 %. Feng et al. [9] proposed a three-dimensional numerical simulation model using the ray tracing method. They obtained the WBCS morphology

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results of 304 stainless steel welded by fiber laser under three states: non-penetration, insufficient penetration, and full penetration. It can be found that different penetration states will affect the dynamics of the keyhole and molten pool, as well as energy absorption, resulting in different weld formations. Attar et al. [10] utilized six different heat source models to simulate the temperature field distribution of 304 stainless steel and copper during disk laser welding, aiming to predict the geometry and size of WBCS. They highlighted the significance of thermal conductivity and specific heat for weld formation.

In the finite element simulation, a heat source was established mainly based on keyhole shape and energy distribution in the keyhole. By continuously adjusting the heat source parameters, the results of the simulated temperature field were compared with experimental results under corresponding parameters. Additionally, the model concerned heat transfer, flow field, velocity field, and other factors, making it more complex. And the mesh division was dense to improve model precision. Consequently, the computation time for most finite element models was relatively lengthy [11,12]. Different from the numerical simulation method, the analytical calculation method reasonably simplified the welding process through a series of assumptions, only retained the more important physical process in the welding process, and solved the welding temperature field through the heat transfer laws. The calculation process was more straightforward, and the calculation time was shorter than the numerical simulation. Lankalapalli et al. [13] established a moving line source with the intensity gradually decreasing in the keyhole depth direction to calculate its corresponding temperature field distribution and obtain the WBCS profile. Steen et al. [14] obtained a WBCS morphology similar to the “nail head” by calculating the temperature field distribution of a point and line combined source. Lampa et al. [15] used a moving linear heat source and realized the prediction of WBCS morphology similar to the “nail head” by increasing the thermal conductivity of the top weld. Ozkat et al. [16] established an analytical model based on energy balance to estimate penetration depth, interface width, and WBCS morphology of lap joints. They revealed that the model could contribute to developing closed-loop machining quality control systems. Giudice et al. [17] established a multipoint-line thermal source model based on heat conduction to simulate the thermal effect during full penetration of high-power CO₂ laser deep penetration welding by adjusting and setting the distribution of laser beam power and heat source to obtain WBCS morphology.

Whether through analytical calculations using the point-line heat source model or finite element numerical simulations to solve temperature field distribution of welding, most studies have assumed that the laser directly interacts with the entire keyhole, undergoing reflections and absorption within the keyhole. Meanwhile, the difference in the influence law of energy coupling law on different welding characteristics needs to be emphasized. Our previous study [18,19] found that the laser directly acted on FKW in high-power fiber LDPW, and the laser energy coupling law within the keyhole involved multiple reflections. However, the keyhole depth was determined by the primary absorption of FKW. Therefore, in this paper, it was considered the difference between the energy coupling law that defines the keyhole depth and front wall morphology or the corresponding energy coupling law that determines the weld formation and proposes a method to calculate WBCS morphology in high-power laser partial penetration welding based on the calculated results of keyhole depth and front wall morphology combined with the effects of multiple reflections.

In this paper, based on the primary absorption of incident laser by FKW, the inclination angle of FKW was calculated point by point according to the balance relationship between laser energy absorbed by the material and the energy required for evaporation at any position of FKW, to obtain the correlation between keyhole depth and FKW morphology with the depth dispersion value. Subsequently, the relationship between absorption and keyhole depth can be established by considering multiple reflections in the keyhole. Furthermore, the mathematical relationship between the fusion boundary profile of weld

cross section and the calculation results of keyhole boundary was calculated based on heat conduction of absorbed energy in the keyhole. Finally, a prediction model of WBCS morphology in LDPW was established by synthesizing the above calculations and verified by welding experiments.

2. Model construction

In LDPW, the focused laser beam with a determined power density distribution acted on the surface of the workpiece with a sufficient thickness, as shown in Fig. 1. The Z-axis was the same as the direction of laser beam. The laser beam moved along the Y-axis with a constant welding velocity. When laser power density exceeds the critical power density of the material, materials in the laser action region will melt and evaporate accordingly and form a keyhole. In the stable state, laser irradiated on FKW and propagated to the keyhole bottom through multiple reflections of FKW, side keyhole wall, and back keyhole wall. Moreover, laser energy absorbed in the keyhole was mainly used to maintain the evaporation and melting of material to form a melt pool, and the melt pool cross-section boundary in the direction vertical to welding velocity was approximated as WBCS morphology.

The primary assumptions in this calculation process are as follows: (1) the welding process was quasi-stable [20]. (2) The calculated FKW morphology and keyhole depth were known (refer to the Appendix A for a detailed calculation process), and the keyhole was presented in a three-dimensional shape of rotational symmetry [21]. (3) The laser energy absorbed in the keyhole was mainly transmitted along the horizontal plane through heat conduction [22], without considering the heat conduction in the depth direction. (4) The effects of metal vapor and particles on beam propagation and energy attenuation were disregarded. (5) The impact of molten pool flow on the calculation results of WBCS was ignored [23]. (6) Heat convection and heat radiation were not considered. During the LDPW, the material at each point along the Y-axis experienced a similar heat transfer process. Consequently, only the propagation of the absorbed laser energy on the XOZ cross-section at the keyhole center along the X-axis was studied, as illustrated in Fig. 2.

2.1. The melting boundary of WBCS at different depth positions

In the calculation process, the laser energy absorbed in the keyhole central section was mainly considered to convert into two parts through heat conduction. The first part is the energy P_v required for maintaining material evaporation in the keyhole area. The other part is the energy P_m required for material melting to form a melt pool. It can be expressed as:

$$P \cdot A = P_v + P_m \quad (1)$$

where P represents laser power, A represents absorptivity. Based on the irregular shape of the keyhole wall, its profile changed with the keyhole depth. Therefore, corresponding to the known profile of the keyhole

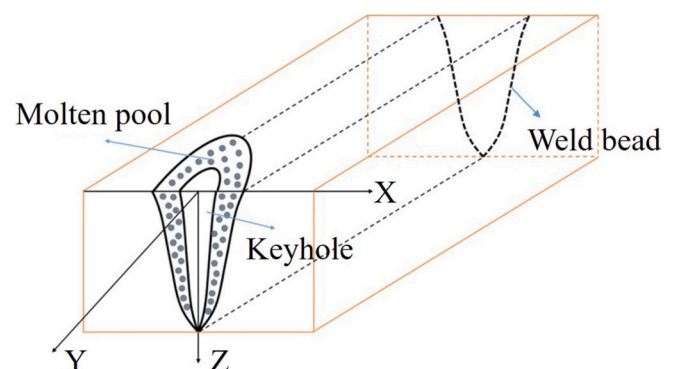


Fig. 1. Schematic of LDPW process.

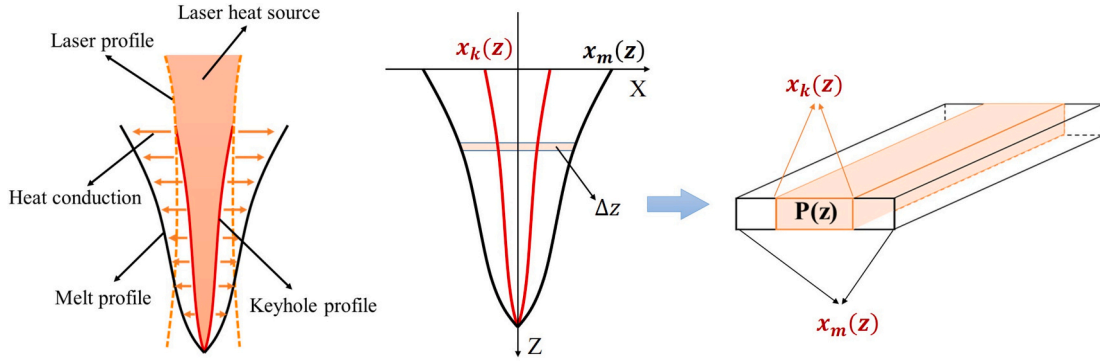


Fig. 2. Schematic of calculation model.

wall, the XOZ section of the keyhole center was micro-differentiated, and a rectangular micro-surface with a length of Δz was arbitrarily selected along the beam action direction, as illustrated in Fig. 2. The width along the X-axis corresponded to the keyhole diameter at the specified depth. The weld was assumed to be symmetrical along the left and right sides of the Z-axis, so the calculation process focused solely on the cross-section profile in the positive direction of the X-axis. The energy required for material evaporation and melting included the energy required for material temperature rising to corresponding temperature and to overcome the latent heat of material itself. The temperature of the melt pool was assumed to be the average of the boiling and melting temperatures. Thus, P_v and P_m corresponding to material in half of the microplane at a certain depth can be expressed as:

$$P_v(z) = x_k(z) \bullet \Delta z \bullet \nu \bullet [\rho_s c_s (T_m - T_0) + L_m \rho_s] + \rho_l c_l (T_v - T_m) \quad (2)$$

$$P_m(z) = x_m(z) \bullet \Delta z \bullet \nu \bullet [\rho_s c_s (T_m - T_0) + L_m \rho_s + \rho_l c_l (T_v - T_m)/2] \quad (3)$$

where $x_k(z)$ and $x_m(z)$ are keyhole and melting boundaries at different depths, respectively. ν is welding velocity. ρ_s and ρ_l represent solid and liquid density, respectively. c_s and c_l represent solid and liquid heat capacity, respectively. L_m represents the latent heat of fusion. T_v represents boiling temperature. T_m represents melting temperature. T_0 represents ambient temperature. Based on the known relation of $x_k(z)$ with keyhole depth and keyhole depth H , assuming that the WBCS was symmetric along the Z-axis and combined with Eqs. (1)–(3), the mathematical relation between melting boundary of WBCS $x_m(z)$ and keyhole boundary $x_k(z)$ at different depth positions can be expressed as follows:

$$x_m(z) = \pm \frac{P \bullet A \bullet \eta - x_k(z) \bullet \Delta z \bullet \nu \bullet \rho_l c_l (T_v - T_m)/2}{\Delta z \bullet \nu \bullet [\rho_s c_s (T_m - T_0) + L_m \rho_s + \rho_l c_l (T_v - T_m)/2]} \quad (4)$$

where η is the proportion of laser energy on the corresponding half of the microplane at different depth positions.

2.2. The proportion of laser energy on each microplane

It was assumed during the calculations that the laser energy propagation was symmetric along the Z-axis, allowing for the calculation of only half of the energy distribution on the microplane. When Δz was sufficiently small, the facet could be approximated as rectangular, enabling the expression of half of its area as follows:

$$S_z = x_k(z) \bullet \Delta z \quad (5)$$

Combined with the total XOZ section area of the keyhole, the proportion η of laser energy on the corresponding half of the microplane at various depths can be expressed as:

$$\eta = \frac{x_k(z) \bullet \Delta z}{\sum_{k=1}^N x_k(z) \bullet \Delta z} \quad (6)$$

where the total number of facets N can be expressed as $H/\Delta z$.

2.3. The estimation of laser absorptivity

For laser absorptivity A in Eq. (1), it was assumed that incident laser energy was absorbed in the keyhole based on multiple reflections. The incident laser was reflected several times in the keyhole before escaping from the keyhole, as illustrated in Fig. 3. The ray tracing method can be employed to obtain the trajectory of its reflected light based on the principles of geometric optics:

$$\vec{R} = \vec{I} + 2(-\vec{I} \bullet \vec{N})\vec{N} \quad (7)$$

where $\vec{N}$, $\vec{I}$ and $\vec{R}$ respectively represent unit vector of normal direction on keyhole wall, incident ray and reflected ray. Taking a two-dimensional axisymmetric cone section with a depth of 5 mm and an opening width of 1 mm as an example, the propagation path of a sectional laser acting vertically in a keyhole was studied, as illustrated in Fig. 3. It can be revealed light density increased with the increase of keyhole depth, indicating that the laser energy absorbed in keyhole, due to multiple reflections on the keyhole wall, was primarily deposited in keyhole bottom. Meanwhile, Allen et al. [24] combined optical coherence imaging and integrating sphere radiation to measure the relationship between absorptivity and keyhole depth, and found that the absorptivity of laser energy at different depth locations was significantly different. Therefore, during the calculation process, the primary absorption at the position on the material surface was about A_p , while the absorption at the keyhole bottom influenced by multiple reflections was about A_r . It was assumed that the absorptivity along the beam action direction was approximately linearly positive correlated with the keyhole depth, which can be expressed as:

$$A = \frac{A_r - A_p}{H} \times z + A_p \quad (8)$$

where H is keyhole depth. Meanwhile, in the calculation process of this paper, A_p was 0.6 [25], and A_r was 0.9 [24].

3. Theoretical calculation results

The MATLAB software can be utilized to solve Eqs. (4), (6) and (8) to obtain the profiles of keyhole and WBCS. A fiber laser and iron were selected for this analysis. Table 1 presents the sectional thermal-physical parameters of materials.

During laser welding, the laser power was set as 4 kW, the welding velocity was set as 2 m/min, the defocusing amount was set as 0, and the focal length was set as 300 mm. The contours of the keyhole and weld boundary on the calculated XOZ plane are exhibited in Fig. 4. The red line represents the keyhole outline, and the black line represents the weld boundary outline. The weld boundary profile and keyhole profile

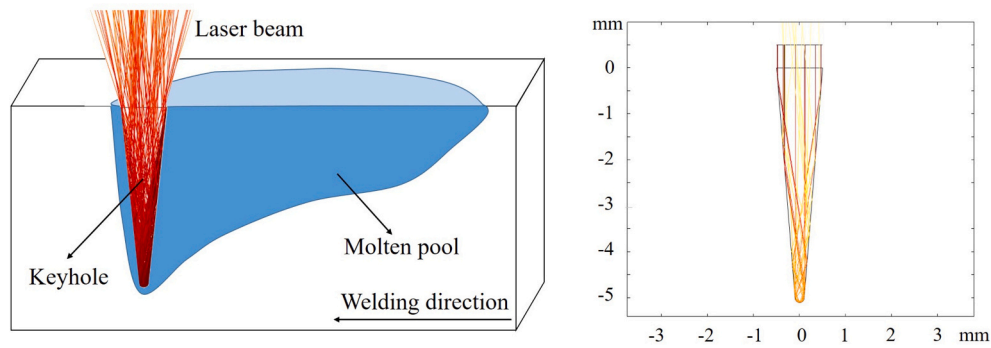


Fig. 3. Schematic diagram of multiple reflections.

Table 1
Sectional thermal-physical parameters of materials.

Property	Symbol	Value
Boiling temperature	T_v	3133 (K)
Melting temperature	T_m	1809 (K)
Liquid density	ρ_l	6518.5 (kg/m ³)
Solid density	ρ_s	7870 (kg/m ³)
Latent heat of vaporization	L_v	6.3639×10^6 (J/kg)
Latent heat of fusion	L_m	2.7196×10^5 (J/kg)
Solid thermal conductivity	λ_s	40.96 (W/(m K))
Liquid thermal conductivity	λ_l	43.99 (W/(m K))
Solid specific heat	c_s	658.63 (J/(kg K))
Liquid specific heat	c_l	804.03 (J/(kg K))

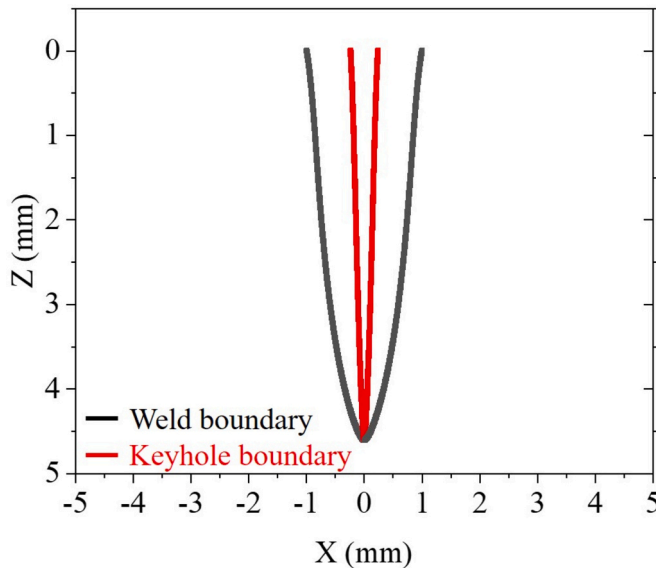


Fig. 4. Boundary contour calculation results of keyhole and weld bead.

were symmetric along the Z-axis, and the morphology of WBCS and keyhole were approximately “V” shape. At the same depth, the distance between the above two outlines along the X-axis decreased with the increase of depth. Since the propagation of energy in the keyhole along the beam action direction was not considered in the prediction model, the calculated weld depth of the cross-section was essentially similar to the keyhole depth.

The welding velocity was set as 1 m/min, the defocusing amount was set as 0, and the laser power was set as 2, 4, and 6 kW, respectively. The morphology of WBCS can be obtained in Fig. 5(a), and the corresponding weld width and depth were calculated as exhibited in Fig. 5(b). From Fig. 5(a), WBCS morphology was generally consistent with the

change of laser power. The area of WBCS increased significantly as the laser power increased. Furthermore, the expanding rate of the cross-section contour remained relatively constant; that means, the difference of the boundary contour at the same depth or the same width was approximately linear with the laser power. Additionally, as indicated in Fig. 5(b), the calculated results for weld width and depth increased approximately linearly with increasing laser power.

The laser power was fixed as 5 kW, the defocusing amount was fixed as 0, and the welding velocity was selected as 2, 4, 6, 8 and 10 m/min, respectively. The calculated morphology of WBCS was exhibited in Fig. 6(a), and the corresponding calculation results of weld width and depth were exhibited in Fig. 6(b). As illustrated in Fig. 6(a), changing velocity had no significant effect on the shape of WBCS, while the cross-section area decreased while increasing welding velocity. Different from the change of WBCS with laser power in Fig. 5(a), increasing welding velocity will gradually slow down the change speed of the cross-section area and boundary profile at the same depth or width position. Additionally, as depicted in Fig. 6(b), weld width and depth decreased with increasing welding velocity, and the rate of decrease gradually slowed.

The laser power was set as 4 kW, the welding velocity was set as 4 m/min, and the defocusing amount was selected to −5, −3, −1, 0, 1, 3, 5 mm, respectively. The calculated morphology of WBCS was exhibited in Fig. 7(a), and the corresponding weld width and depth were calculated in Fig. 7(b). From Fig. 7(b), when the defocusing amount increased in a positive direction, weld width increased, and depth decreased; consequently, the ratio of weld depth to width decreased accordingly, which also corresponded to the changes in the shape of WBCS. In contrast to the positive defocusing amount, in the negative defocusing amount, the weld width decreased first and then increased, while the weld depth increased first and then decreased when the absolute defocusing amount increased. Furthermore, from Fig. 7(b), when the absolute value of defocusing amount was the same, compared with the negative defocusing amount, the weld width and depth calculated under the positive defocusing amount condition were relatively larger and smaller. Under the above parameter conditions, when the defocusing amount was −1 mm, the calculated weld depth was maximum, and the weld width was minimum.

4. Experimental verification

4.1. Experimental setup

A fiber laser with a Rayleigh length of 3.08 mm and a spot diameter of 0.3 mm was utilized. A CO₂ with a Rayleigh length of 3.45 mm and a spot diameter of 0.27 mm was used. The laser beam was applied vertically to the plate surface for bead-on-plate scanning welding. To minimize the influence of plume or plasma during laser welding, a supersonic transverse gas curtain was positioned 5 mm above the plate surface, with the airflow directed perpendicular to the welding direction. The schematic of the experimental device is illustrated in Fig. 8. Finally,

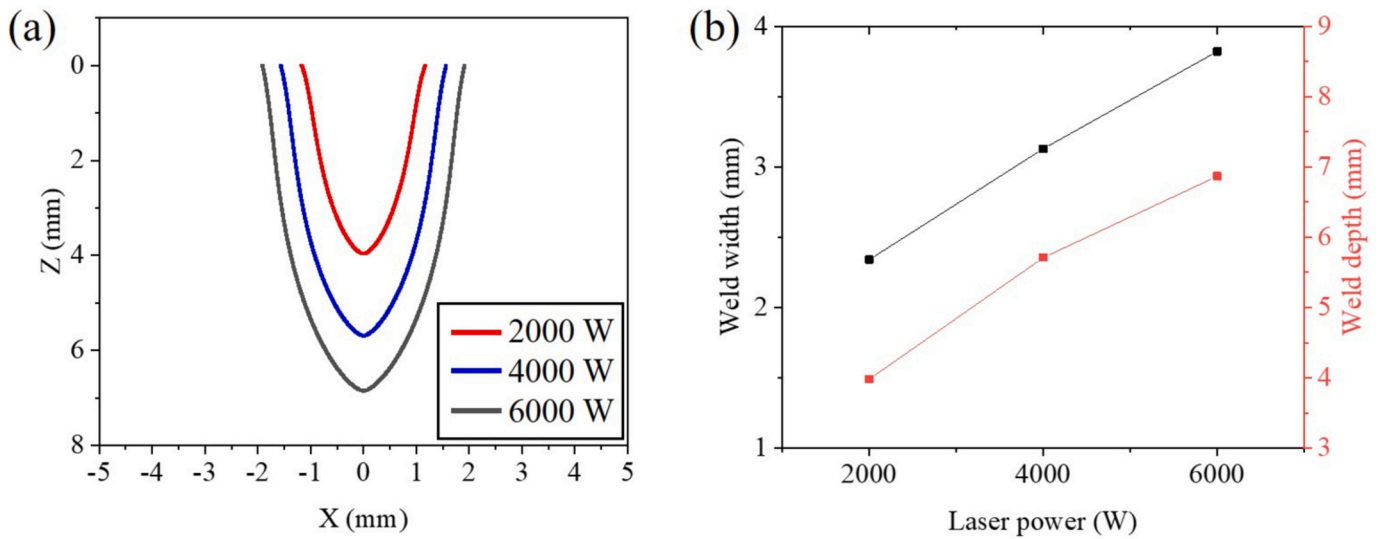


Fig. 5. The variation of WBCS morphology (a), weld depth, and width (b) with laser power.

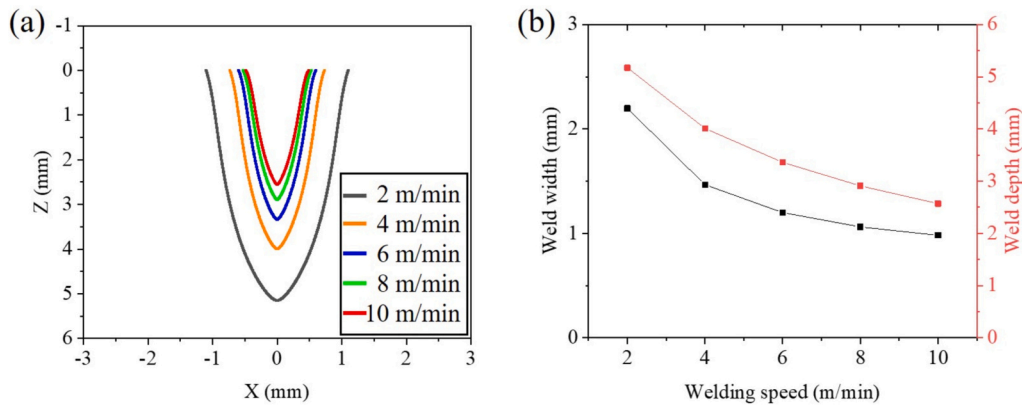


Fig. 6. The variation of WBCS morphology (a), weld depth, and width (b) with welding velocity.

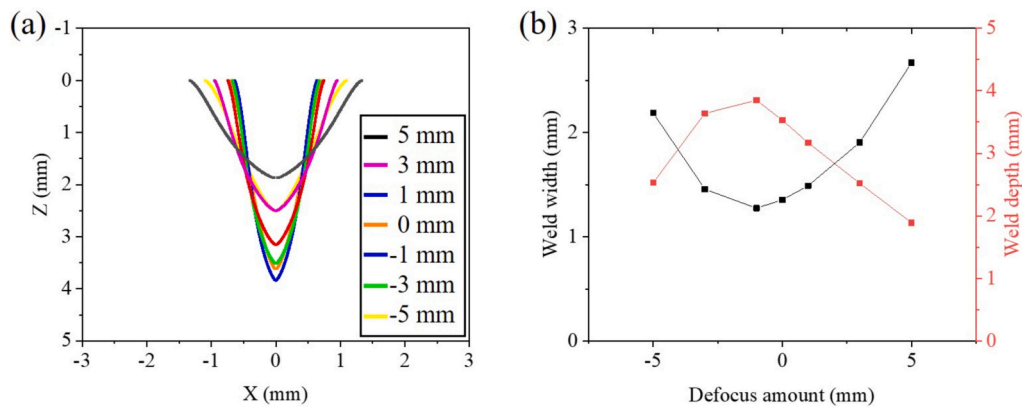


Fig. 7. The variation of WBCS morphology (a), weld depth, and weld width (b) with the defocusing amount.

WBCS morphology can be obtained through metallographic processing and microscope.

4.2. Experimental verification

The YLS-6000 fiber laser was selected for welding, with the welding velocity fixed at 2 m/min and the laser power set at 5 kW. The

experimental measurement results and calculation results (outlined in red) of WBCS morphology are presented in Fig. 9(a). Subsequently, the laser power was adjusted to 3 kW, and the welding velocity maintained at 2 m/min and 4 m/min, respectively. The experimental and calculated results are shown in Fig. 9(b)-(c). From the figures, it can be observed that the calculated WBCS exhibits a “V” shape, whereas the experimental WBCS resembles a “wine glass” shape. At the upper part of the

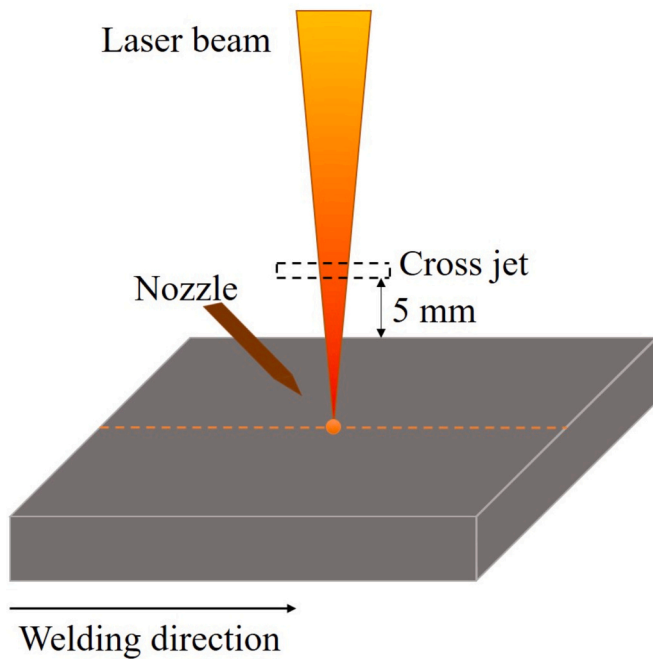


Fig. 8. Schematic of the experimental device.

weld, the calculated weld width was smaller than the experimental result, and the arc of the boundary contour was different to some extent. Near the weld bottom, the calculated contour of the weld was similar to

the tip of the keyhole, while the experimental contour of the weld was relatively gentle. Near the center height of the weld, the calculated weld profile aligns closely with the experimental results.

Afterward, the fiber laser and CO₂ laser were selected for welding. The laser power was set to 2.8 kW, and the welding velocities were chosen at 1, 3, 5 and 7 m/min, respectively. A comparison of the calculated and measured results regarding weld width and depth is presented in Figs. 10–11. From the figure, weld width and depth from calculations decreased while increasing velocity, and their variation rules were basically consistent with experimental results. However, under the same parameter conditions, there were still some differences between the calculated and experimental results. In most cases, the calculated values for weld width and depth were smaller than the experimental measurements.

5. Analysis and discussion

In the process of LDPW, as shown in Fig. 12, the laser directly interacts with FKW, and the keyhole depth is determined by the absorption of the incident laser by FKW. Consequently, the keyhole wall profile and keyhole depth can be obtained by studying the interaction between front wall and incident laser based on the above rules (Appendix A). After that, the reflected light corresponding to incident light acting on FKW continues to interact with keyhole wall. The repeated reflection and absorption of the laser in keyhole lead to a redistribution of laser energy, ultimately influencing the final weld formation. The actual distribution of beam energy in the keyhole, resulting from multiple reflections, is closely related to the keyhole shape. Additionally, during the process of multiple reflections, the influence of the first and second absorptions on the total energy absorption is particularly significant

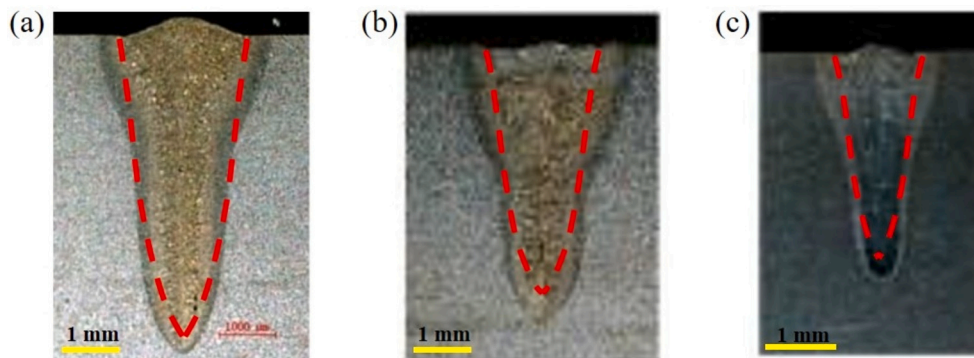


Fig. 9. Comparison of calculated and experimental results about WBCS morphology.

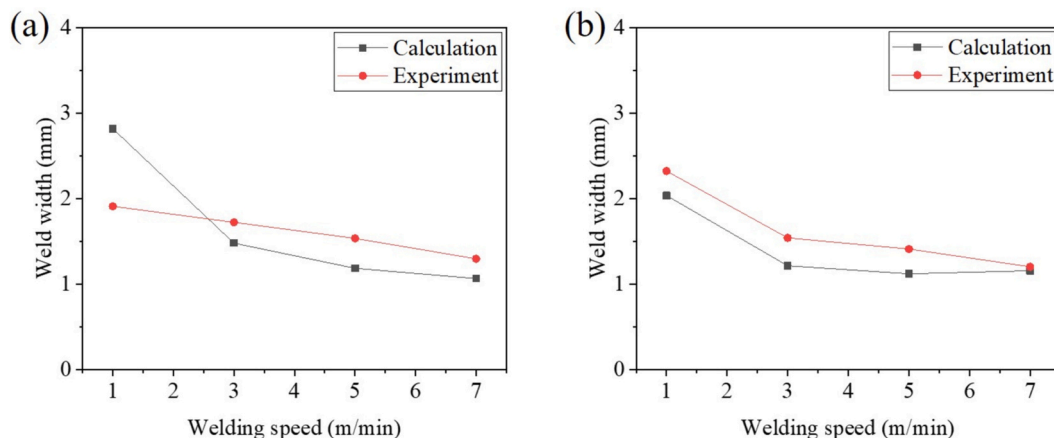


Fig. 10. Weld width results from calculated and experimental for fiber laser (a) and CO₂ laser (b) welding as a function of welding velocity.

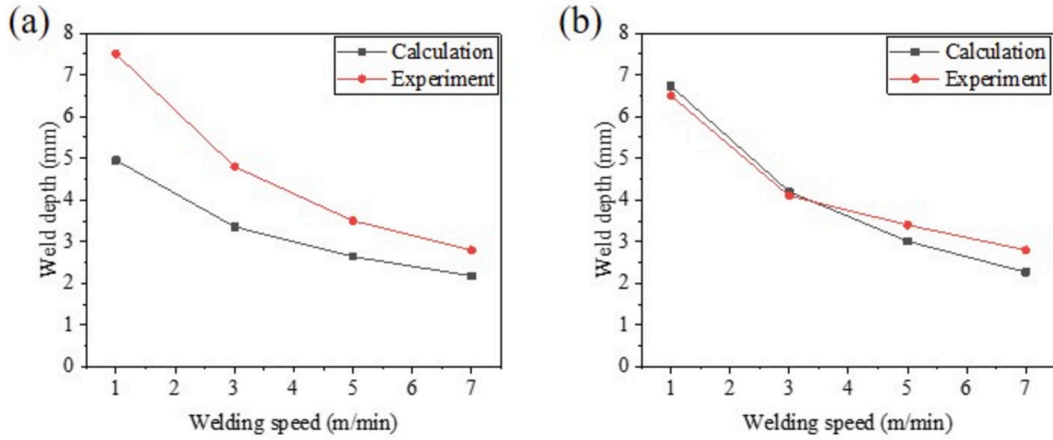


Fig. 11. Weld depth results from calculated and experimental for fiber laser (a) and CO₂ laser (b) welding as a function of welding velocity.

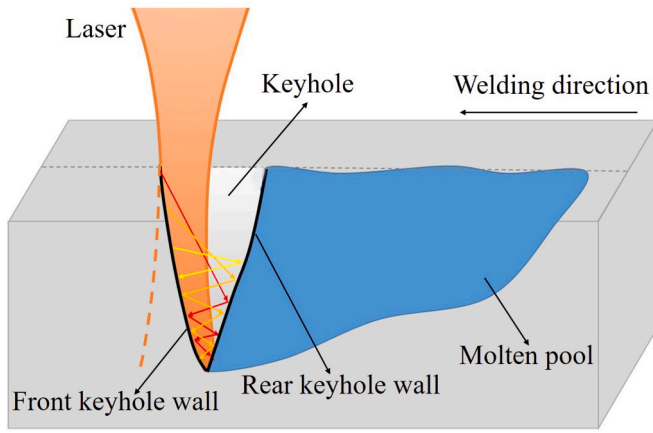


Fig. 12. Schematic diagram of LDPW.

[26]. The keyhole shape is approximately conical, with a small opening range and considerable depth. Therefore, the light reflected from the front wall predominantly impacts on the keyhole bottom, causing the laser energy to be deposited mainly at this location. However, in practical LDPW, the specific energy distribution in the keyhole is highly complex, influenced by numerous factors [27]. Therefore, during the calculation process, the absorptivity was assumed to increase linearly with the change of keyhole depth. This assumption regarding absorptivity differs from the actual welding process, thereby affecting the accuracy of the calculation results.

In the calculation process, only the heat conduction effect of absorbed laser energy in the keyhole on the horizontal plane along the direction parallel to welding velocity was considered. This means that the weld formation around the keyhole at the same depth was the only factor taken into account. However, in the actual process, the laser energy absorbed in keyhole will also propagate along the beam action direction. Consequently, this portion of the laser energy gradually propagates downward, leading to a downward broadening of the weld boundary profile. This phenomenon resulted in the calculated width of weld profile near the weld bottom, as shown in Fig. 10, being smaller than the experimental results. Additionally, the influence of high temperature at keyhole bottom on weld depth, due to thermal diffusion, was ignored. Assuming that the normal vector of the keyhole contour boundary at the bottom position is parallel to the Z-axis, and considering only the thermal diffusion at this point along the positive direction of the Z-axis, it can be expressed as [28]:

$$\frac{\partial T}{\partial t} = \frac{\lambda_l}{C_l} \frac{\partial^2 T}{\partial z^2} = h_l \frac{\partial^2 T}{\partial z^2} \quad (9)$$

where h_l is the thermal diffusion coefficient of the liquid material. According to Eq. (9), the actual weld depth should be increased in the Z-axis positive direction, which aligns with the weld depth calculation results shown in Fig. 11, indicating that the calculated values are generally smaller than the experimental measurements.

To simplify the calculations to a certain extent, the model presented in this paper considers only heat conduction, neglecting the effect of molten pool flow on weld formation. In actual LDPW, the effect of molten pool flow on weld formation is primarily manifested through the Marangoni effect, which is induced by the surface tension gradient in the molten pool and the effect of metal vapor eruption in the keyhole [29]. In the upper region of the molten pool, the temperature decreases as the distance from the keyhole region increases. As the temperature of the molten pool decreases, the surface tension coefficient increases, and the Marangoni force resulting from the surface tension gradient can be expressed as follows:

$$\tau = \frac{d\gamma}{dT} \frac{dT}{dx} \quad (10)$$

where γ represents surface tension coefficient and T represents molten pool temperature. As illustrated in Fig. 13, the liquid metal in the upper

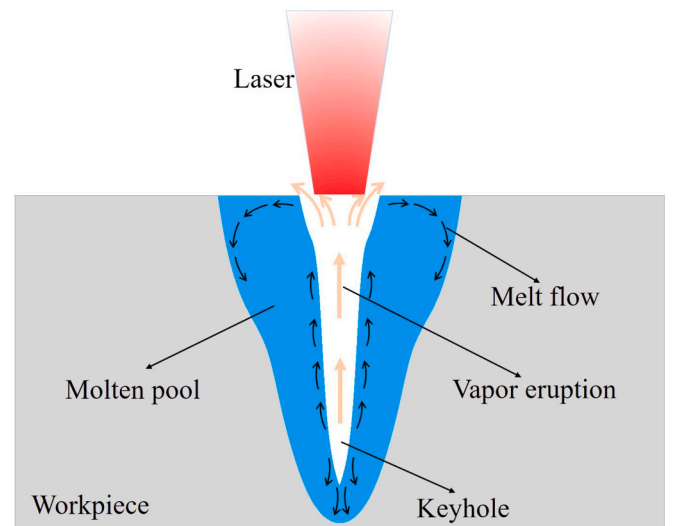


Fig. 13. Schematic diagram of molten pool flow.

region of molten pool flows away from the keyhole area due to the action of Marangoni force, resulting in a gradual decrease in the corresponding temperature gradient until it reaches an inflection point. At that time, the flow direction of the molten metal will reverse. It is precisely the flow of molten metal in the upper part of weld pool, moving from the keyhole to the sides, driven by the Marangoni effect, that accounts for the discrepancies observed in the morphology of the top of WBCS between the calculated and experimental measurement results (Fig. 9). This effect also contributes to the calculated weld width smaller under most conditions (Fig. 10).

Keyhole is the primary feature of LDPW. The vapor recoil pressure in the keyhole, along with the friction between the vapor erupting from the keyhole and the keyhole wall, causes the liquid metal near the keyhole bottom to flow upward. The vapor recoil pressure can be expressed as follows [30]:

$$P_r = AB_0 \sqrt{T_s} \exp(-U_e/k_b T_s) \quad (11)$$

where U_e represents latent heat of evaporation of each atom, k_b represents Boltzmann constant, A and B_0 represent evaporation constant, and T_s represents surface temperature of keyhole. The friction force between metal vapor and keyhole wall can be represented as [31]:

$$\tau_g = 4\eta_g \frac{V_g}{r_0} \quad (12)$$

where η_g represents the dynamic viscosity of metal vapor, V_g represents the average velocity of vapor eruption, and r_0 represents keyhole radius. Under the combined influence of the aforementioned pressures, gravity, and fluid pressure, a flow characteristic in the molten pool is observed, similar to that depicted Fig. 13. As illustrated in Fig. 13, the molten metal at keyhole bottom flows diagonally downward under the recoil pressure, which results in a weld depth obtained from experiments that is greater than that predicted by calculations. Simultaneously, the molten metal is more likely to accumulate at the keyhole bottom [32]. When the laser moves, the molten metal cools and solidifies, causing the shape of WBCS bottom to become more rounded. However, during the calculation process, the cooling and solidification of molten pool are not considered, leading to a shape of WBCS bottom that is more similar to the tip shape.

In the welding process, the plume and plasma significantly influence weld formation. The beam is attenuated as it passes through the plume and plasma, and the attenuation rate can be represented by the Lambert-Beer law as follows:

$$\alpha = 1 - e^{-KI} \quad (13)$$

Where K is the attenuation coefficient of plume or plasma to the laser. Both the plume and plasma not only attenuate laser energy but also induce a heating effect on the material surface under certain conditions. This heating can increase the weld surface width and change the cross-section shape. Furthermore, the weld shape produced by laser welding in a vacuum environment tends to resemble a “V” shape [33]. Additionally, the plume and plasma influence laser propagation, thereby affecting the shape of laser spot and the laser energy distribution on the material surface [34]. Although a transverse gas curtain was used to blow away the plume or plasma during experiments, it was difficult to eliminate its effect on welding process, especially the metal vapor nearly keyhole aperture was difficult to remove. Consequently, neglecting the plume or plasma in calculations can lead to discrepancies between the results of calculations and experimental outcomes. Moreover, heat convection and heat radiation were not considered in the calculation process, which may overestimate the absorption of laser energy to some extent, thereby compromising the accuracy of calculation results.

Considering the numerical simulation method, the coupling of multiple physical fields can consider the aforementioned aspects as comprehensively as possible. However, the process mainly verifies

model accuracy by calculating WBCS, comparing with experimental results, and modifying model parameters accordingly. Based on this, the distribution patterns of other physical fields, such as the flow and velocity fields, can be analyzed. It is important to note that high computational accuracy often correlates with extended calculation times. For instance, Courtois et al. [35] required approximately 16 h to simulate 10 milliseconds of the resolution process. Zhou et al. [36] took 6 h to simulate about 100 milliseconds of welding. Mayi et al. [37] introduced an innovative methodology to model the “beam trapping” effect in the multiphysical modeling of laser welding. For three-dimensional models, the computation may take 1–5 days or even longer (exceeding 1 month) after the model stabilizes. However, the model proposed in this paper is specifically designed for calculating the boundary profile of WBCS, and the corresponding geometric parameters such as weld depth and width can be obtained according to the profile. It allows for the determination of geometric parameters such as weld depth and width based on the profile, without focusing on the specific temperature distribution and flow dynamics within the weld pool. When the laser wavelength, focusing system parameters, welding process parameters, and welding materials are known, the calculation results of WBCS morphology can be obtained directly in a relatively short time (approximately 3–4 s) based on the mathematical model established in this study. Furthermore, a direct and definitive correlation between the calculated results of WBCS morphology and the relevant parameters has been established, eliminating the need to adjust heat source parameters based on engineering experience or experimental results. Additionally, the absolute error between the calculated results of weld depth and width in this model and the experimental measurement results is approximately 0.5–1 mm, with a relative error of about 5–20 %. These results generally meet the requirements for rapid computation.

In fact, the model establishment requires further improvement and expansion in the future. Firstly, regarding the differences in WBCS morphology illustrated in Fig. 9, the influence of molten pool flow, especially the recoil pressure and the Marangoni effect, should be considered. Additionally, with the development of computer technology, numerical calculation methods such as the finite volume method and finite element method can be applied [38]. This makes estimating the discontinuous change of material temperature more accurate under the action of three-dimensional heat conduction and the thermal physical parameters of materials that change with temperature. However, the application of numerical methods may increase computation time; thus, finding a balance between calculation accuracy and computational efficiency remains a key focus in current simulation and calculation efforts. Furthermore, the dynamic fluctuations of the plume/plasma (including volume, temperature, density, and characteristics of the particles in the plume) complicate the accurate quantification of its impact on laser processing. Meanwhile, the application of machine learning methods in laser processing can also provide a new direction for processing complex nonlinear problems and improving model accuracy [39]. Finally, regarding the practical application of the model, it is essential to consider a wider range of actual processing conditions and materials.

6. Conclusions

In this paper, the boundary profile of the keyhole was constructed by considering only the front wall, and the keyhole depth was calculated using the primary absorption of the front wall. Considering the multiple reflection and absorption effects of laser beam in the keyhole, a mathematical relationship between the fusion boundary profile of WBCS and the keyhole boundary was established. This relationship allows for the prediction of WBCS morphology during LDPW by analyzing the energy balance in the keyhole central section, oriented vertically to the welding velocity, based on the principles of heat conduction. The following conclusions can be drawn:

- (1) The model calculation results indicate that the variations in weld geometric characteristics (weld depth and width) and WBCS morphology with the laser power, welding velocity, defocusing amount, and other parameters are generally consistent with established consensus. A comparison between the model calculations and experimental measurements under certain parameters further demonstrates the model's feasibility.
- (2) The results of comparing and verifying the model indicate that the influence of the energy coupling law on keyhole depth (weld depth) and WBCS morphology (weld formation) is different. The keyhole depth is mainly determined by the primary absorption of FKW, while WBCS morphology is closely associated with the multiple reflections occurring within the keyhole.
- (3) There are still discrepancies between the WBCS morphology results obtained from calculations and experiments, which exhibit a “V” shape and a “wine cup” shape, respectively. Under most parameters, the calculated results of weld width and depth are smaller than the experimental results. These discrepancies are mainly attributed to the omission of several factors, including the Marangoni effect induced by temperature gradient at the upper part of the molten pool, the effect of recoil pressure, the accumulation and solidification of the melt at the keyhole bottom, and the heating effect of plume near the molten pool surface.
- (4) The absolute error between the calculated weld depth and width in the model and the experimental measurements is approximately 0.5 to 1 mm, while the relative error ranges from 5 % to 20 %. The computation time is around 3 to 4 s. Future

enhancements and extensions of the model could be developed by incorporating factors such as molten pool flow, numerical computations, plume/plasma dynamics, machine learning techniques, and multi-dimensional validation.

CRediT authorship contribution statement

Baoqi Zhu: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation. **Wuxiong Yang:** Methodology, Data curation, Conceptualization. **Jianglin Zou:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Conceptualization. **Shun Xie:** Supervision, Methodology, Data curation. **Tao Liu:** Validation, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The calculation of FKW morphology and keyhole depth

In the calculation process, only the profile of FKW on the plane parallel to welding velocity was considered, and the inclination angle of each point on FKW was calculated based on the energy balance relationship of FKW, which can be expressed as:

$$\tan\theta = \frac{q_v}{I \cdot A} \quad (A1)$$

where I represents incident laser energy, A represents absorptivity, and q_v represents the energy required for material evaporation. The incident-focused laser beam has a rotationally symmetric Gaussian power density distribution, which can be expressed as:

$$I(r, z) = \frac{2P}{\pi\delta(z)^2} \exp\left(-\frac{2r^2}{\delta(z)^2}\right) \quad (A2)$$

where P is the laser power and $\delta(z)$ is the radius of the laser spot at different depth positions, which can be expressed as:

$$\delta(z) = \sqrt{\frac{4K_f^2 f^2}{D^2} + \frac{D^2}{4f^2} (z + \Delta F)^2} \quad (A3)$$

where D represents diameter of laser spot before focusing, f represents focal length, K_f represents value of beam focusing characteristic parameter, and ΔF represents defocusing amount. The energy required for material evaporation q_v included the energy required for the material temperature to rise from ambient temperature to boiling temperature under the action of heat conduction and the energy absorbed by the material to overcome its latent heat during the melting process, which can be expressed as:

$$q_v = (T_v - T_0) \lambda P e' \left(\cos\varphi + \frac{K_1(Pe'r)}{K_0(Pe'r)} \right) + \rho_s L_m \cdot v \quad (A4)$$

where (r, φ) represents the position coordinates with the line source as the origin, λ represents thermal conductivity, $Pe' = v/2\kappa$ represents Peclet number, v represents welding velocity, K_0 and K_1 represent modified Bessel function of the zeroth and first order, respectively. Then, by considering the discrete value Δz of z_i from the top of FKW on the workpiece surface according to the inclination angle of each point on FKW, the position coordinate of front wall contour x_f was obtained:

$$x_{f,i+1} = x_{f,i} - \Delta z \cdot \tan\theta_i \quad (A5)$$

When the inclination angle of a point at FKW bottom is equal to $\pi/4$ ($\tan\theta = 1$), the complete morphology of FKW can be obtained, and the keyhole depth can be obtained accordingly.

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